

CHAPTER 4

Facilities Location and Layout

4.1. Introduction to Facilities Location

A *facility* is something built or established to serve a purpose. *Facilities management* is a location decision for that facility and the composition or internal layout of the facility once located. This chapter discusses the location and layout of a facility. A sample location problem which is used in subsequent discussions is given in Example 4.1.

Example 4.1 Location of a Manufacturing Facility

Management of the Waterstill Manufacturing Company has decided to embark upon a manufacturing facility expansion. The expansion is necessary because of an increase in sales of its product—water softeners. There are currently two facilities, one in Tulsa, Oklahoma, and the other in Tempe, Arizona, but there is no additional space available at either site. Furthermore, the company has warehouses in Washington, D.C., Cleveland, Ohio; Lincoln, Nebraska; San Francisco, California; and Phoenix, Arizona. They feel that transportation costs might be reduced if a new plant at a different location were constructed. Figure 4.1 shows the existing sites.

Current sales suggest the need for a 10% plant expansion, but forecasts over the next 10 years indicate a steady increase up to double the current sales volume. Then the volume is expected to level off. Because of the relatively unskilled labor requirement and desired

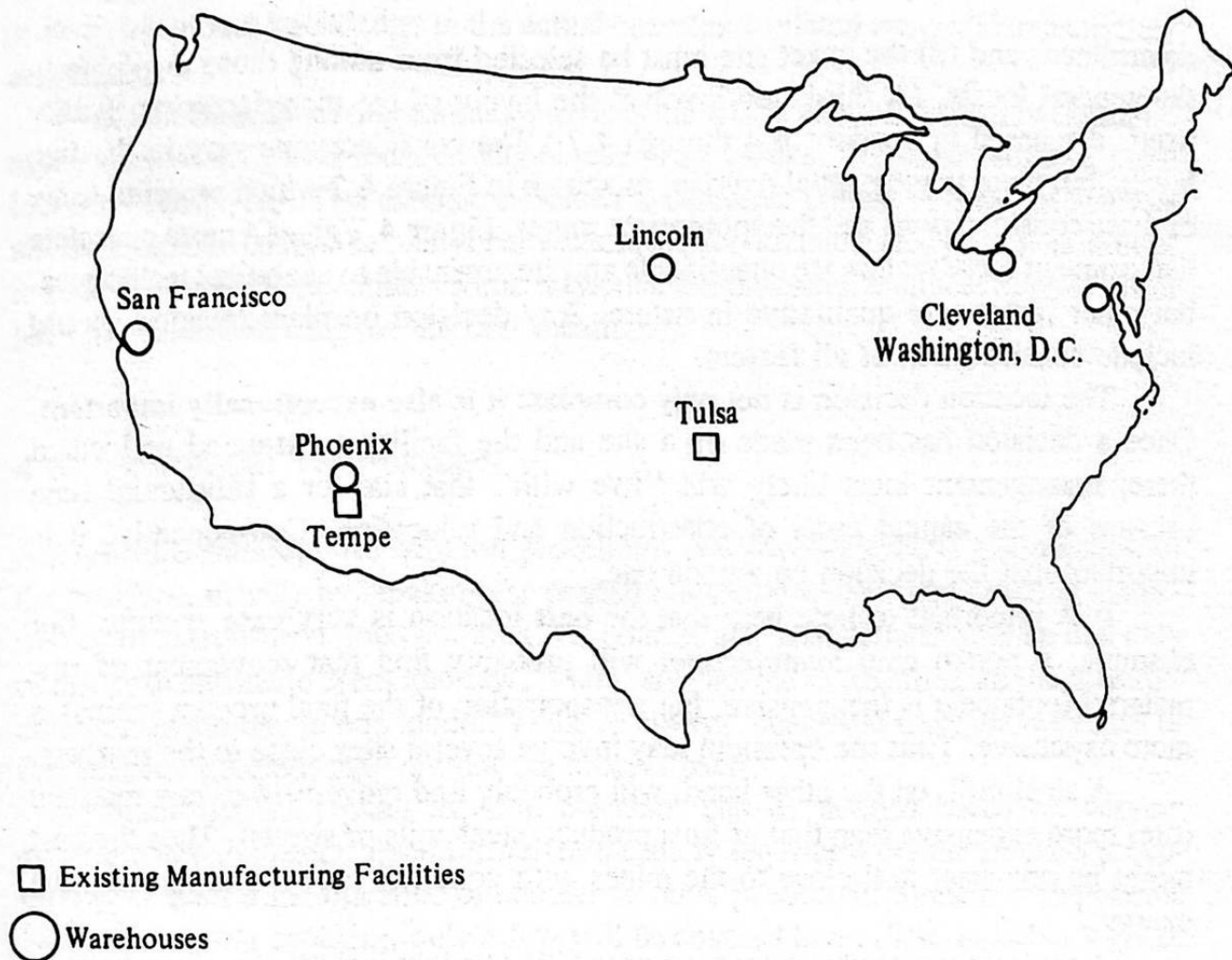


Fig. 4.1. Waterstill Company—manufacturing and wholesale facilities.

stability of the labor force, a rural location is preferred for the new manufacturing facility. As implied by the warehouses, the market is distributed evenly throughout the country and raw materials availability is no problem. Transportation costs are proportional to the distance traveled, regardless of warehouse location, and it is a company policy to be able to supply any warehouse within 1 week of receipt of an order.

The Waterstill Manufacturing Company has hired your company, the Whitehurst Cowboy Consulting Firm, to aid them in their search for a suitable location. Much information is supplied above, but the company is completely willing to provide any additional information you feel is necessary. Your job is to determine the best location for the new plant.

4.2. Considerations

Although the information provided in the example is certainly not enough to locate the new facility, it does give an idea of the immense complexity of such an undertaking. The problem is so immense and so complex that there is no one “best way” of carrying on such a study. There are analytical procedures to aid in the process (these will be discussed in Section 4.3), but the decision must also include factors that are difficult if not impossible to quantify.

The decision usually is made in two stages: (1) the general location must be

determined, and (2) the exact site must be selected from among those available in the general locale. (A third step involves the layout of the manufacturing facility itself, discussed in Sections 4.4 through 4.7.) The considerations vary in the two levels, but there is substantial overlap, as shown in Figure 4.2 which presents some of these considerations and the appropriate stages. Figure 4.7 gives a more complete list. Some of these factors are quantifiable and are amenable to analytical techniques, but other factors are qualitative in nature. Any decision on plant location should include consideration of all factors.

The location decision is not only complex; it is also exceptionally important. Once a decision has been made on a site and the facility constructed or located there, management most likely will "live with" that site for a substantial time because of the capital costs of construction and relocation. Consequently, it is important that the decision be a good one.

It is important to note here that the best location is very case specific. For example, a potato chip manufacturer will probably find that movement of raw material (potatoes) is inexpensive, but transportation of the final product (chips) is more expensive. Thus the optimum may involve several sites close to the markets.

A steel mill, on the other hand, will probably find movement of raw material (ore) more expensive than that of final product (steel coils or sheets). Thus the best might be one steel mill close to the mines with good rail access and inexpensive energy.

A typical team studying location possibilities might include accountants, lawyers, marketing experts, various consultants, executives, and industrial engineers. With this entourage of talent, the industrial engineer usually is not asked to be knowledgeable in all areas. Instead, the industrial engineer may be called upon to

<i>Location Factor</i>	<i>Territory Selection</i>	<i>Selection of Site and Community</i>
1. Market	*	
2. Raw Materials	*	
3. Transportation	*	*
4. Power	*	*
5. Climate and Fuel	*	
6. Labor and Wages	*	*
7. Laws and Taxation	*	*
8. Community Services and Attitude		*
9. Water and Waste		*

Fig. 4.2. Consideration in the two stages of plant location. [From *Plant Layout and Design* © James M. Moore, 1962. Reprinted with permission of the publisher. (New York: Macmillan Publishing Company, 1962.)]

provide talent and knowledge in the actual operations related areas and in analytical techniques.

Industrial engineering training provides the I.E. with a solid math background that includes analytical location techniques. This chapter discusses some of these location techniques. The chapter is neither a complete course in location analysis nor a complete course in analytical techniques for location analysis. It is a brief introduction that demonstrates some ways that the industrial engineer applies his or her talents in searching for the best location.

4.3. Analytical Techniques

Analytical techniques are solution procedures that involve careful examination of the problem, usually by breaking the problem into components and utilizing mathematics to search for a good solution. Of course, any analytical technique can only consider quantifiable areas and few, if any, are capable of handling all the quantifiable considerations in one model. Thus, as always, analytical tools are only aids in the decision process; they are not the process itself.

Many different plant location problems can be defined based on varying criteria and parameters. Each formulation usually requires a unique solution procedure or at least a modification of another solution procedure. Since it is impossible to mention every problem, only a few will be covered here. First, possible ways of stratifying plant location problems are mentioned.

Usually, the criterion or objective is to minimize some cost function. Often, the distance traveled is chosen as the cost function. Thus, the objective becomes one of minimizing total distance traveled. Sometimes the objective, or at least a consideration, may be to minimize the maximum distance traveled (as, for example, in locating fire stations or hospitals where the maximum or worst response distance is important).

Another possible stratification is in the distance measure used. Sometimes the straight-line or Euclidean distance may be more appropriate. If two facilities are located at points represented by (X_1, Y_1) and (X_2, Y_2) , then the Euclidean distance between the two is

$$[(X_1 - X_2)^2 + (Y_1 - Y_2)^2]^{1/2}$$

while the rectilinear distance is

$$|X_1 - X_2| + |Y_1 - Y_2|$$

The two are demonstrated in Figure 4.3. A rectilinear measure is often used when the streets are laid out in grids, as in an urban setting (Philadelphia, Pennsylvania, in Figure 4.3). An Euclidean measure is used when a straight-line distance is more appropriate, as in interstate or intercity travel (Columbus to Akron in Figure 4.3). Still another classification scheme is on the number of facilities to be located—single vs. multiple. For example, the location of one fire station in Manhattan,

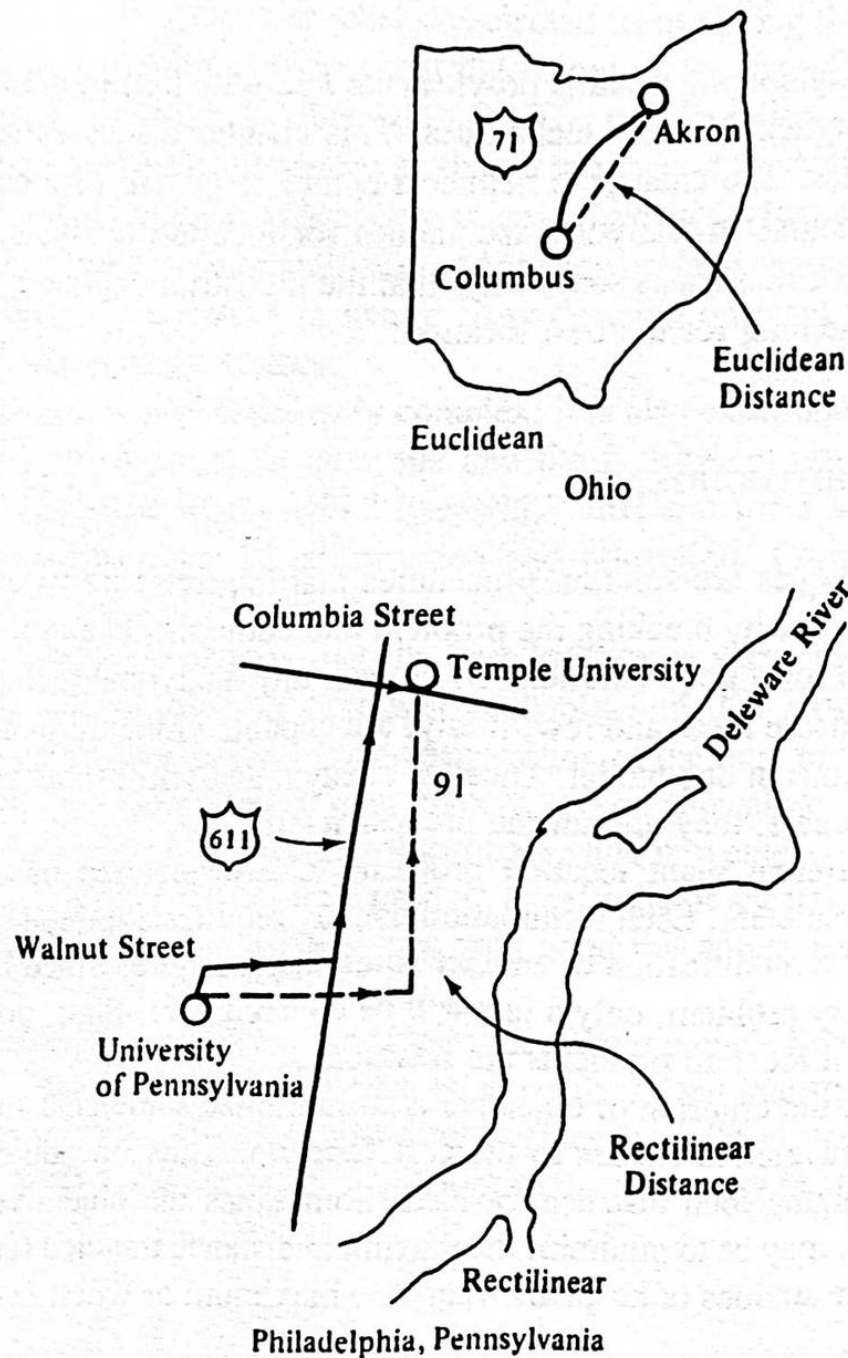


Fig. 4.3. Euclidean and rectilinear distance examples.

Kansas, may require a different technique from the location of ten fire stations in Oklahoma City, Oklahoma.¹

4.3.1 Transportation Method of Linear Programming

It is relatively simple to formulate a manufacturing plant location problem as a transportation linear programming problem involving minimization of total cost to distribute the product. (Linear programming will be discussed in depth in Section 14.8.) In this case, the transportation linear programming formulation is used to determine the best distribution pattern for the plant in a certain location. Then, total

¹For amplification, see Richard L. Francis and John A. White, *Facility Layout and Location: An Analytical Approach* (Englewood Cliffs, N.J.: Prentice Hall, 1974), pp. 21–26.

cost can be calculated, various locations tried, and the location with the least total cost chosen.

For example, suppose that the Waterstill Manufacturing Company has narrowed its choice of locations to two—Youngstown, Ohio, and Sparta, South Carolina. Assume further that the costs of manufacturing and distribution to the various warehouses can be calculated as shown in Table 4.1, in which information was obtained through careful examination of forecasts and production and distribution costs.

To formulate this as a transportation linear programming problem, it first must be realized that there are two alternatives: (1) locate the plant in Youngstown or (2) locate the plant in Sparta. There will, therefore, be two transportation problems, one with the plant in Sparta and one with the plant in Youngstown.

Consider the cost of production and distribution. If the Tulsa plant produced and shipped a softener to Washington, the total cost would be

$$\begin{array}{rcc} \$75.00 & \$5.00 & \$80.00 \\ \text{Production} + \text{Distribution} & = & \text{Total Cost per unit} \end{array}$$

The cost in the transportation matrix would then be \$80.00 for the Tulsa–Washington combination. For Tulsa–Cleveland, it would be

$$\$75.00 + \$3.00 = \$78.00$$

while for Tempe–Lincoln, it would be

$$\$70.00 + \$3.50 = \$73.50$$

If we continue this for all combinations and remember that the plants cannot exceed capacity and that the warehouses do not need more than their demand, the transportation problem, with the plant in Youngstown, can be formulated as in Figure 4.4.

Table 4.1. PRODUCTION AND DISTRIBUTION COSTS PER WATER SOFTENER

<i>To</i> <i>From</i>	<i>Wash- ington</i>	<i>Cleve- land</i>	<i>Lin- coln</i>	<i>San Fran- cisco</i>	<i>Phoe- nix</i>	<i>Weekly Capac- ity</i>	<i>Produc- tion Costs</i>
Tulsa	5.00	3.00	2.00	3.00	2.00	7,000	\$75.00
Tempe	6.50	5.00	3.50	1.50	.20	5,500	\$70.00
Youngstown	1.50	.50	1.80	6.50	5.00	12,500	\$70.00
Sparta	3.80	5.00	8.00	7.50	8.00	12,500	\$67.00
Forecasted Weekly Demand	5,000	6,000	4,000	7,000	3,000		

From \ To	Washington	Cleveland	Lincoln	San Francisco	Phoenix	Plant Capacity
Tulsa	80.00	78.00	77.00	78.00	77.00	7,000
Tempe	76.50	75.00	73.50	71.50	70.20	5,500
Youngstown	71.50	70.50	71.80	76.50	75.00	12,500
Demand (Weekly)	5,000	6,000	4,000	7,000	3,000	Total 25,000

Fig. 4.4. Transportation formulation—plant in Youngstown.

Here the two existing plants are shown as sources along with the potential plant in Youngstown.

The number in the upper right-hand corner of each square indicates the total cost of production and distribution per unit for that combination. The numbers under plant capacity are the production capability for each plant and the numbers in the row for demand are the demand at the warehouses. The total capacity equals total demand; this must be true for the procedure to work. If total demand does not equal total capacity, an artificial plant (warehouse) with zero cost and the necessary capacity (demand) can be used to achieve the required balance.²

To record a solution, numbers must be added to the various blocks to indicate the number of units involved in that transfer. For example,

From \ To	Washington
Tulsa	500 80.00

implies that 500 units are being produced in Tulsa and shipped to Washington at a total cost of $500 \text{ units} \times \$80.00/\text{unit} = \$40,000$.

There are several ways of solving this problem. Some give optimal solutions with a relatively high cost of time and effort and others yield very fast, good but not necessarily optimal solutions with a relatively low cost. The latter procedures are called *heuristic*; the former procedures are called *optimal seeking*. An optimal seeking method is described in Section 14.8.2, but for this chapter a heuristic method that quickly yields a good solution is presented. This method is the *least cost assignment* routine.

²This will be discussed in further detail in Chapter 14.

The first step in the least cost assignment procedure is to pick the least costly element and assign as much as possible to that combination remembering that demand or capacity cannot be exceeded. Then the demand and capacity are corrected to show partial assignment and the procedure continues. For example, the Tempe-Phoenix combination is the least expensive at \$70.20 per unit. Tempe can produce 5,500 units, but Phoenix only needs 3,000, so 3,000 units should be assigned to that combination. Now, Phoenix has 0 units demand and Tempe has 2,500 units capacity. Because Phoenix does not need any more water softeners, that column can be dropped from consideration. The next cheapest is Youngstown-Cleveland at \$70.50 per unit. Six thousand units can be assigned there and the Cleveland column can be dropped from consideration. If we continue in this manner, the final solution would be as shown in Figure 4.5, in which the circled numbers imply the order in which the combinations were chosen.³

The total cost of this solution (which may or may not be optimal) is

$$2,500 (77) + 4,500 (78) + 2,500 (71.50) + 3,000 (70.20) + 5,000 (71.50) \\ + 6,000 (70.50) + 1,500 (71.80) = \$1,821,050$$

The corresponding formulation and solution for the plant located in Sparta are given in Figure 4.6. The total cost of this solution (which again may or may not be optimal) is

$$4,000 (77) + 3,000 (78) + 2,500 (71.50) + 3,000 (70.2) + 5,000 (70.80) \\ + 6,000 (72) + 1,500 (74.50) = \$1,829,100$$

Both solutions (Figures 4.5 and 4.6) were obtained using heuristic rather than optimal seeking methods. Thus the solutions are probably "close to optimum" rather than optimum. In Section 14.8.2 we take the solution in Figure 4.6 and show how to obtain the true optimum.

The total weekly cost of production and shipment of Youngstown is \$1,821,050, which is \$8,050 per week *more economical* than the total cost of production and shipment at Sparta. This implies that if all other factors are equal, the plant should be built in Youngstown.

All other factors are probably not equal, however. For example, the wages have been accounted for in the transportation model, but the labor market has not. Is the needed labor available? Are the specialized trades available? What is the attitude of local labor? All of these are questions that must be answered before a decision can be made. The costs developed in Figures 4.5 and 4.6 must be considered along with all the other factors in the proper perspective.

If the entries in the matrix of Figure 4.4 were profit instead of cost, then maximum profit assignment should be used. That is, each assignment is made to

³In step 3 either the Tempe-San Francisco or the Youngstown-Washington combination can be chosen since both are the next cheapest. The Youngstown-Washington combination was chosen since 5,000 units can be assigned there and only 3,000 can be assigned to the Tempe-Phoenix combination. This is one way of breaking ties.

From \ To	Washington	Cleveland	Lincoln	San Francisco	Phoenix	Plant Capacity
Tulsa	80.00	78.00	⑥ 77.00 2,500	⑦ 78.00 4,500	77.00	7,000
Tempe	76.50	75.00	73.50	④ 71.50 2,500	① 70.20 3,000	5,500
Youngstown	③ 71.50 5,000	② 70.50 6,000	⑤ 71.80 1,500	76.50	75.00	12,500
Demand (Weekly)	5,000	6,000	4,000	7,000	3,000	25,000

Fig. 4.5. Solution to transportation problem using least cost assignment—Youngstown.

the remaining feasible combinations with maximum profit. Capacities and demands are adjusted as before and the procedure is continued.

4.3.2 Multiple Objectives

The preceding section demonstrated that there are many factors to be considered in the final location decision. One technique for analytically considering many factors is the use of a method for decision making with multiple objectives. The method, which is very similar to the point rating plan of job evaluation (to be discussed in Chapter 6), is needed because so many factors, such as local climate, local attitude, fire and police protection, and so on, cannot be readily quantified. Yet they are important in the location decision.

As an example, suppose that the Waterstill Manufacturing Company has narrowed its choice down to two locations, city A and city B. All cost calculations

From \ To	Washington	Cleveland	Lincoln	San Francisco	Phoenix	Plant Capacity
Tulsa	80.00	78.00	⑥ 77.00 4,000	⑦ 78.00 3,000	77.00	7,000
Tempe	76.50	75.00	73.50	③ 71.50 2,500	① 70.20 3,000	5,500
Sparta	② 70.80 5,000	④ 72.00 6,000	75.00	⑤ 74.50 1,500	75.00	12,500
Demand (Annual)	5,000	6,000	4,000	7,000	3,000	25,000

Fig. 4.6. Solution to transportation problem using least cost assignment—Sparta.

have been made and there is no clear-cut distinction. In fact, for simplicity, assume that all costs are equal at the two locations. How can the decision be made?

The first step is to make a list of all important factors. There are several lists available—Reed gives a list of 21 noncost factors and Moore⁴ gives a list of 36. The list from Reed is reproduced in Figure 4.7.

It is important that the list contain all noncost factors exerting an influence on the decision to insure that no significant items are omitted.

The next step is to assign relative point values for each of the factors for the specific company and plant to be located. The values may vary somewhat between companies and within one company between different plants. As an example, the values shown in Figure 4.8 are suggested by Reed as maximum for the 21 factors listed in Figure 4.7. This is not to imply that all companies should use these values; but they are presented as an example.

If we assume that the Waterstill Company uses this scheme, the next step is to assign degrees and points within each factor. Usually, from 4 to 6 degrees are used with a linear assignment of points between degrees. As an example, Figure 4.9 shows the degrees for factor 16 (community attitude) (this list is drawn from Reed, whose book develops degrees and points for each factor).

At this point Waterstill has its evaluation scheme completely defined, so it now must assign each of the two locations (A and B) degrees and corresponding points for each factor. The methods used to obtain the degrees for each location vary substantially. Usually, a team of experts does the rating and management has the right to veto or change the results. Figure 4.10 shows hypothetical results. In factor 16 Waterstill has decided that city A has a cooperative labor force (degree 3), but city B has a noncooperative labor force (degree 2).

Waterstill and the Whitehurst Cowboy Consulting Firm now compare these results with the cost calculations and make a decision. City A has a total point value of 719 compared to 643 for city B. City A would probably be preferred since all cost calculations were assumed equal. If they had not been, some trade-off in the decision process would have to occur.

4.3.3 Mathematical Programming (Optional)

Much work has recently been done toward optimally determining the number, location, and sizes of plants (warehouses) through what is known as *mathematical programming*. Mathematical programming consists of representing the cost of a system by a mathematical expression, constraining the realm of possible solutions when necessary, and searching over the realm of the solutions to find the optimum point (minimum or maximum).

In the case of plant location, an expression specifying the cost of the system is generated and constraints are incorporated to insure feasibility. In the model to

⁴James M. Moore, *Plant Layout and Design* (New York: Macmillan Publishing Company, 1962), pp. 48–58.

- | | |
|--------------------------------|---|
| (1) Nearness to Market | (12) Churches and Religious Facilities |
| (2) Nearness to Unworked Goods | (13) Recreational Opportunities |
| (3) Availability of Power | (14) Housing |
| (4) Climate | (15) Vulnerability to Air Attack |
| (5) Availability of Water | (16) Community Attitude |
| (6) Capital Availability | (17) Local Ordinances |
| (7) Momentum of Early Start | (18) Labor Laws |
| (8) Fire Protection | (19) Future Growth of Community |
| (9) Police Protection | (20) Medical Facilities |
| (10) Schools and Colleges | (21) Employee Transportation Facilities |
| (11) Union Activity | |

Fig. 4.7. List of 21 noncost factors in plant location. [Extracted with permission from Rudell Reed, Jr., *Plant Layout: Factors, Principles, and Techniques* (Homewood, Ill.: Richard D. Irwin, Inc., 1961), p. 361.]

<i>Factor—Value</i>	<i>Factor—Value</i>	<i>Factor—Value</i>
1—280	8—10	15—10
2—220	9—20	16—60
3—30	10—20	17—50
4—40	11—60	18—30
5—10	12—10	19—30
6—60	13—20	20—10
7—10	14—10	21—20

Fig. 4.8. Maximum point values for each factor.

<i>Degrees</i>		<i>Point Assignment</i>
0	Hostile, Bitter, Noncooperative	0
1	Parasitic in Nature	15
2	Noncooperative	30
3	Cooperative	45
Maximum	Friendly and More Than Cooperative	60

Fig. 4.9. Degress and points for factor 16 (community attitude).

Factor	CITY A		CITY B	
	Degree	Points	Degree	Points
1	Maximum	280	3	168
2	4	176	4	176
3	2	12	4	24
4	0	0	4	24
5	4	8	2	4
6	3	36	4	48
7	2	4	1	2
8	Maximum	10	2	4
9	4	16	2	8
10	2	8	3	12
11	3	36	3	36
12	2	6	2	6
13	3	15	Maximum	20
14	4	8	0	0
15	1	2	2	5
16	3	45	2	30
17	2	20	4	40
18	3	23	1	8
19	0	0	3	18
20	1	2	1	2
21	4	12	2	8
Total		719		643

Fig. 4.10. Hypothetical results on plant location.

be described a number of potential plant sites are first determined. Locating a plant at a particular site involves a fixed cost (construction, insurance) and a variable cost of supplying customers. The model, when solved, yields the optimum number, location, and size of the plant. Chapter 14 will discuss mathematical programming solution procedures. This chapter only presents the formulation of the problem.

The following is taken from Efroymsen and Ray,⁵ but many other authors treat similar problems.

The following notation is necessary:

m = number of customers

n = number of potential plant sites (determined in advance)

k_j = fixed cost of opening plant j

x_{ij} = percent or fraction of customer i 's demand that is satisfied by factory j

⁵M. A. Efroymsen and T. L. Ray, "A Branch and Bound Algorithm for Plant Location," *Operations Research*, 14 (1966), 361-368.

$$y_j = \begin{cases} 0 & \text{if a plant is not located at } j \\ 1 & \text{if a plant is located at } j \end{cases}$$

c_{ij} = cost of supplying the total demand of customer i by plant j

If all of customer i 's demand were supplied by plant j , the variable cost of shipment would be just c_{ij} . Since fractional deliveries are allowed, the total variable cost for the amount customer i receives from plant j becomes $c_{ij}x_{ij}$, and the total variable cost for all customers and plants is given by

$$\sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \quad (4.1)$$

$\left(\sum_{i=1}^m \right)$ is a symbol signifying "the summation of m elements in the order of 1 through m ."

If a plant is located at site j , a fixed cost of k_j is incurred. If not, no cost is incurred. This can be formulated as $k_j y_j$ since y_j is 1 only if a plant is located at site j and 0 otherwise. The total fixed costs of locating all plants then are given by

$$\sum_{j=1}^n k_j y_j \quad (4.2)$$

The total cost is the sum of Equations (4.1) and (4.2), which yields

$$\text{total cost } TC = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} + \sum_{j=1}^n k_j y_j \quad (4.3)$$

Some constraints are necessary. First, a customer cannot be supplied from j unless a plant is located at j , and the total amount supplied by a plant located at j cannot exceed the total demand. This is represented by Equation (4.4) (remember that x_{ij} is a fraction and y_j is either 0 or 1). The lowercase letter, n , at the right-hand side of the expression indicates the total number of inequalities (i.e., $j = 1, \dots, n$ implies that there are n expressions where j takes the values 1 through n . $\sum_{i=1}^m x_{i1} \geq my_1$; $\sum_{i=1}^m x_{i2} \geq my_2$; and so on).

$$\sum_{i=1}^m x_{ij} \leq my_j \quad (j = 1, \dots, n) \quad (4.4)$$

Next, all customers must be supplied. This can be represented by

$$\sum_{j=1}^n x_{ij} = 1, \quad (i = 1, \dots, m) \quad (4.5)$$

Finally, all x_{ij} must be 0 or positive and all y_j must be 0 or 1. These can be represented by

$$y_j = (0,1) \quad (j = 1, \dots, n) \quad (4.6)$$

$$x_{ij} \geq 0 \quad (i = 1, \dots, m \text{ and } j = 1, \dots, n) \quad (4.7)$$

Putting all these into one model yields

$$\text{minimize } TC = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} + \sum_{j=1}^n k_j y_j$$

subject to

$$\begin{aligned} \sum_{i=1}^m x_{ij} &\leq m y_j \quad (j = 1, \dots, n) \\ \sum_{j=1}^n x_{ij} &= 1 \quad (i = 1, \dots, m) \\ y_j &= (0,1) \quad \forall j \\ x_{ij} &\geq 0 \quad \forall i, j \end{aligned}$$

$\forall j$ simply means "for each j " or "for every j " and $\forall ij$ means "for each i and j ." The term *subject to* means that the realm of possible solutions is restricted by the constraints that follow.

Since the equation or objective function is for total cost, the goal is to minimize it subject to the constraints. This can be done by using the techniques of linear programming and branch and bound search, but they will not be covered here. Instead, an example is given.

Example 4.2

Suppose that the Whitehurst Cowboy Consulting Firm decided to use such a model in its efforts to aid another company, Indeng Enterprises. This company wants to start a new product line; so it has to construct one or more manufacturing plants capable of supplying its four warehouses located in Norfolk, Virginia; Chicago, Illinois; Seattle, Washington; and Baton Rouge, Louisiana. The company has determined three potential sites. They are in Baton Rouge, Louisiana; Cookeville, Tennessee; and Kansas City, Kansas. The situation is shown in Figure 4.11.

To aid Indeng Enterprises, the Whitehurst Cowboys would first have to determine some cost information at the three potential sites. Specifically, the cost of supplying the demand at any location by a plant must be determined. That information is shown in Table 4.2.

It is relatively simple to develop the mathematical formulation when given the information shown in Table 4.2. For example, x_{11} is the percentage of total goods supplied to warehouse 1 from potential plant 1. Therefore, the cost coefficient in the objective function is the cost of supplying warehouse 1 from plant 1 or \$85,000. Similarly, the cost of supplying warehouse 1 from potential plant 2 is \$75,000. The first two terms in the cost expression are then

$$85,000 x_{11} + 75,000 x_{12}$$

and the total variable cost is

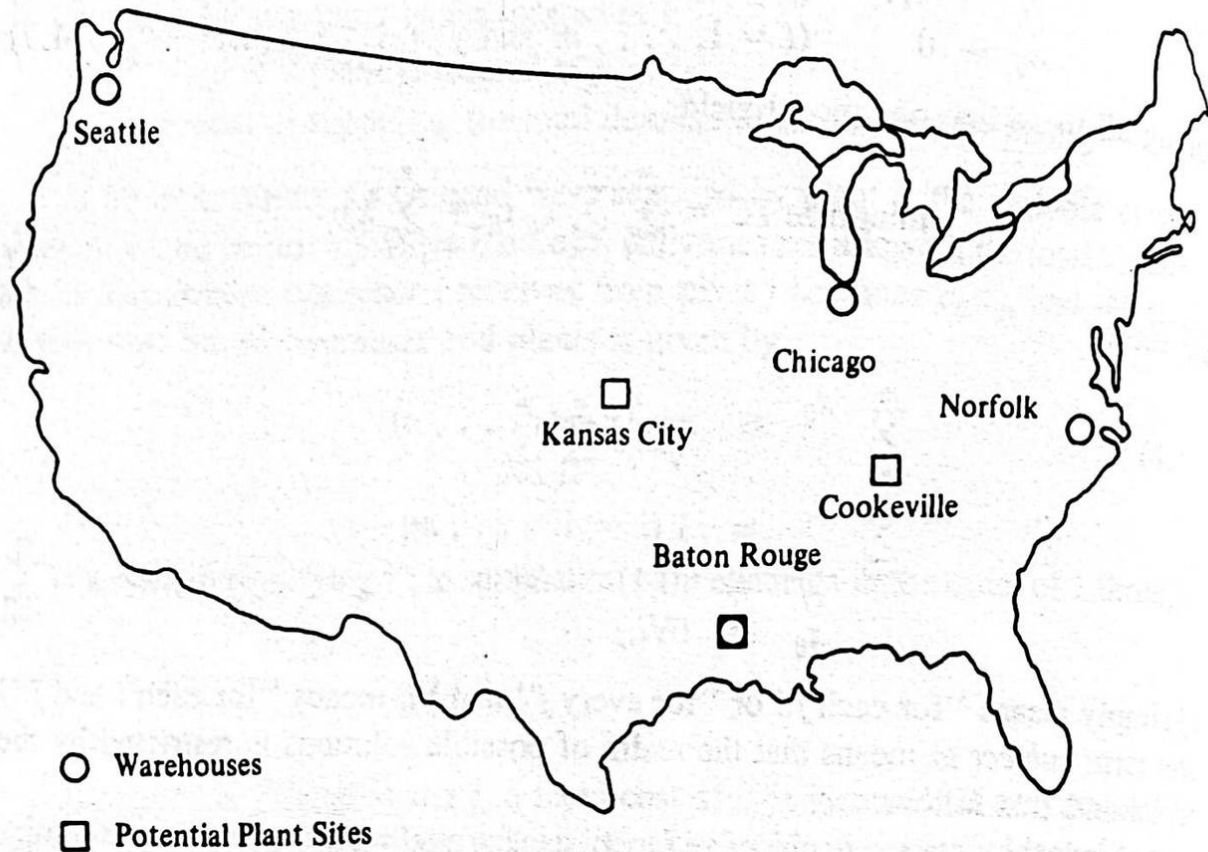


Fig. 4.11. Indeng Enterprises warehouses and potential plant sites.

$$\begin{aligned}
 &85,000 x_{11} + 75,000 x_{12} + 100,000 x_{13} + 95,000 x_{21} \\
 &+ 85,000 x_{22} + 40,000 x_{23} + 200,000 x_{31} \\
 &+ 185,000 x_{32} + 185,000 x_{33} + 10,000 x_{41} \\
 &+ 70,000 x_{42} + 60,000 x_{43}
 \end{aligned}$$

Table 4.2. COST INFORMATION FOR INDENG ENTERPRISES

From \ To	Potential Plant Site	Baton Rouge	Cookeville	Kansas City
Warehouse Site		(1)	(2)	(3)
Norfolk (1)		85,000	75,000	100,000
Chicago (2)		95,000	85,000	40,000
Seattle (3)		200,000	185,000	185,000
Baton Rouge (4)		10,000	70,000	60,000
Fixed Cost to Open Plant		60,000	50,000	65,000

Since the fixed cost associated with opening plant 1 is \$60,000, the first term in the expression for fixed cost is \$60,000 y_1 and the total fixed cost is

$$\$60,000 y_1 + \$50,000 y_2 + \$65,000 y_3$$

The total cost is the sum of fixed cost and variable cost or

$$\begin{aligned} &85,000 x_{11} + 75,000 x_{12} + 100,000 x_{13} + 95,000 x_{21} + 85,000 x_{22} \\ &+ 40,000 x_{23} + 200,000 x_{31} + 185,000 x_{32} + 185,000 x_{33} \\ &+ 10,000 x_{41} + 70,000 x_{42} + 60,000 x_{43} + 60,000 y_1 \\ &+ 50,000 y_2 + 65,000 y_3 \end{aligned}$$

The first set of m constraints ($m = 3$) states that the total amount shipped from any plant cannot exceed the total demand or that a plant must be open (i.e., $y_j = 1$) before any shipment can occur. Those constraints are

$$\begin{aligned} x_{11} + x_{21} + x_{31} + x_{41} &\leq 4y_1 \\ x_{12} + x_{22} + x_{32} + x_{42} &\leq 4y_2 \\ x_{13} + x_{23} + x_{33} + x_{43} &\leq 4y_3 \end{aligned}$$

The next set of constraints requires that all demands must be satisfied.

$$\begin{aligned} x_{11} + x_{12} + x_{13} &= 1 \\ x_{21} + x_{22} + x_{23} &= 1 \\ x_{31} + x_{32} + x_{33} &= 1 \\ x_{41} + x_{42} + x_{43} &= 1 \end{aligned}$$

Finally, the 0, 1 and non-negativity constraints are

$$\begin{array}{lllll} y_1 = 0, 1 & x_{11} \geq 0 & x_{21} \geq 0 & x_{31} \geq 0 & x_{41} \geq 0 \\ y_2 = 0, 1 & x_{12} \geq 0 & x_{22} \geq 0 & x_{32} \geq 0 & x_{42} \geq 0 \\ y_3 = 0, 1 & x_{13} \geq 0 & x_{23} \geq 0 & x_{33} \geq 0 & x_{43} \geq 0 \end{array}$$

Putting all this together yields

$$\begin{aligned} \text{Minimize } TC &= 85,000 x_{11} + 75,000 x_{12} + 100,000 x_{13} \\ &+ 95,000 x_{21} + 85,000 x_{22} + 40,000 x_{23} \\ &+ 200,000 x_{31} + 185,000 x_{32} + 185,000 x_{33} \\ &+ 10,000 x_{41} + 70,000 x_{42} + 60,000 x_{43} \\ &+ 60,000 y_1 + 50,000 y_2 + 65,000 y_3 \end{aligned}$$

subject to

$$\begin{aligned} x_{11} + x_{21} + x_{31} + x_{41} &\leq 4y_1 \\ x_{12} + x_{22} + x_{32} + x_{42} &\leq 4y_2 \\ x_{13} + x_{23} + x_{33} + x_{43} &\leq 4y_3 \\ x_{11} + x_{12} + x_{13} &= 1 \\ x_{21} + x_{22} + x_{23} &= 1 \\ x_{31} + x_{32} + x_{33} &= 1 \\ x_{41} + x_{42} + x_{43} &= 1 \end{aligned}$$

$$y_1 = 0, 1, \quad y_2 = 0, 1, \quad y_3 = 0, 1$$

$$\begin{aligned} x_{11} \geq 0, \quad x_{12} \geq 0, \quad x_{13} \geq 0, \quad x_{21} \geq 0, \quad x_{22} \geq 0, \quad x_{23} \geq 0 \\ x_{31} \geq 0, \quad x_{32} \geq 0, \quad x_{33} \geq 0, \quad x_{41} \geq 0, \quad x_{42} \geq 0, \quad x_{43} \geq 0 \end{aligned}$$

The solution procedure for this formulation is simple, but development of it is not really necessary for understanding the model. Those interested should consult Efroymson and Ray (footnote 5).

The optimal solution to this problem obtained through the use of the solution procedure or exhaustive enumeration in this simple problem is

$$\begin{aligned} y_1^* &= 1, \quad y_2^* = 0, \quad y_3^* = 1 \\ x_{11}^* &= 1, \quad x_{23}^* = 1, \quad x_{33}^* = 1, \quad x_{41}^* = 1 \\ x_{12}^* &= x_{13}^* = x_{21}^* = x_{22}^* = x_{31}^* = x_{32}^* = x_{42}^* = x_{43}^* = 0 \end{aligned}$$

This says that plants are to be constructed in locations 1 and 3. Warehouses 1 and 4 are to be serviced by plant 1, and warehouses 2 and 3 are to be serviced by plant 3. According to this model, Indeng Enterprises should then locate plants in Baton Rouge, Louisiana, and in Kansas City, Kansas.

4.3.4 Public-Sector Location Problems

There has been much interest recently in the use of industrial engineering in the location of public-sector service agencies. Fire stations, hospitals, emergency centers, equipment yards, and landfills are just a few that have been discussed. Often, the criterion is different from that for private-sector problems and the constraints may differ. The objective in the private sector is usually (and has been in all examples to this point) to minimize some total cost function often represented by time or distance. The distance measure is frequently Euclidean.

In the public sector the objective is frequently to minimize some maximum response time, for example, in time to get to a fire or emergency. Also, since almost all problems are in an urban setting, the distance measure is often rectilinear. Of course, there may be problems in the private sector whose objectives are to minimize some maximum response time with a rectilinear measure and vice versa, but, in general, the foregoing observations hold.

A typical public-sector location problem is given in Example 4.3. It will be analyzed and solved later.

Example 4.3

The Whitehurst Cowboy Consulting Firm has been contacted by the American Red Cross of Oklahoma City, Oklahoma. A new blood bank is to be constructed somewhere in the city that will serve 15 existing hospitals and research centers. A given number of trips weekly are to be made by a truck from the blood bank to a given hospital or center and then back to the blood bank. The number of trips between the blood bank and any given facility

is relatively constant and known, but the number varies between facilities. The Red Cross wants to locate the blood bank such that the *total distance* traveled by the truck is minimized. The details, including map coordinates and the number of trips weekly, are shown in Table 4.3. Figure 4.12 is a map showing approximate locations of all hospitals and research centers.

The problem described in Example 4.3 is a single-facility location problem with 15 existing facilities. The objective given by the Red Cross is to minimize the total distance traveled by the truck. Since the problem occurs in an urban setting, the distance measure is most likely rectilinear (which it is in this case).

This location problem is one of the easiest to solve. An algorithm for the solution is given below. The proof is not shown, but Francis and White (footnote 1) give a complete proof and discussion of the more complicated problems.

Solution Procedure. Location of a new facility with respect to several existing facilities (rectilinear distances assumed):

- (1) Arrange all facilities in increasing order of the X coordinates. Add up the number of trips for that same order. The optimum X location is such that no more than half the trips are to the left and no more than half to the right.
- (2) Repeat for the Y coordinate such that no more than half the trips are above and no more than half below.
- (3) The X and Y coordinates just chosen are the optimum locations. The price of land, neighborhood, and other parameters must now be considered.

Table 4.3. INFORMATION FOR THE BLOOD BANK LOCATION STUDY

<i>Hospital or Research Center</i>	<i>Map Location</i>	<i>Number of Trips Weekly</i>
1. Hillcrest	(22.5, 5.5)	10
2. South Community	(26.5, 9.5)	10
3. Capitol Hill	(29.5, 14.0)	5
4. St. Anthony	(28.5, 24.0)	3
5. Bone and Joint	(28.5, 24.5)	1
6. Mercy	(29.0, 25.0)	3
7. Polyclinic	(30.0, 25.0)	1
8. Wesley	(29.5, 24.5)	1
9. University	(32.0, 24.5)	3
10. University School of Medicine	(32.0, 25.0)	1
11. Oklahoma Medical Research Institute	(32.5, 25.0)	1
12. Veteran's Administration Hospital	(33.0, 25.0)	1
13. Crippled Children's	(32.5, 24.5)	1
14. Deaconess	(16.0, 36.5)	5
15. Baptist Memorial	(18.0, 37.5)	4

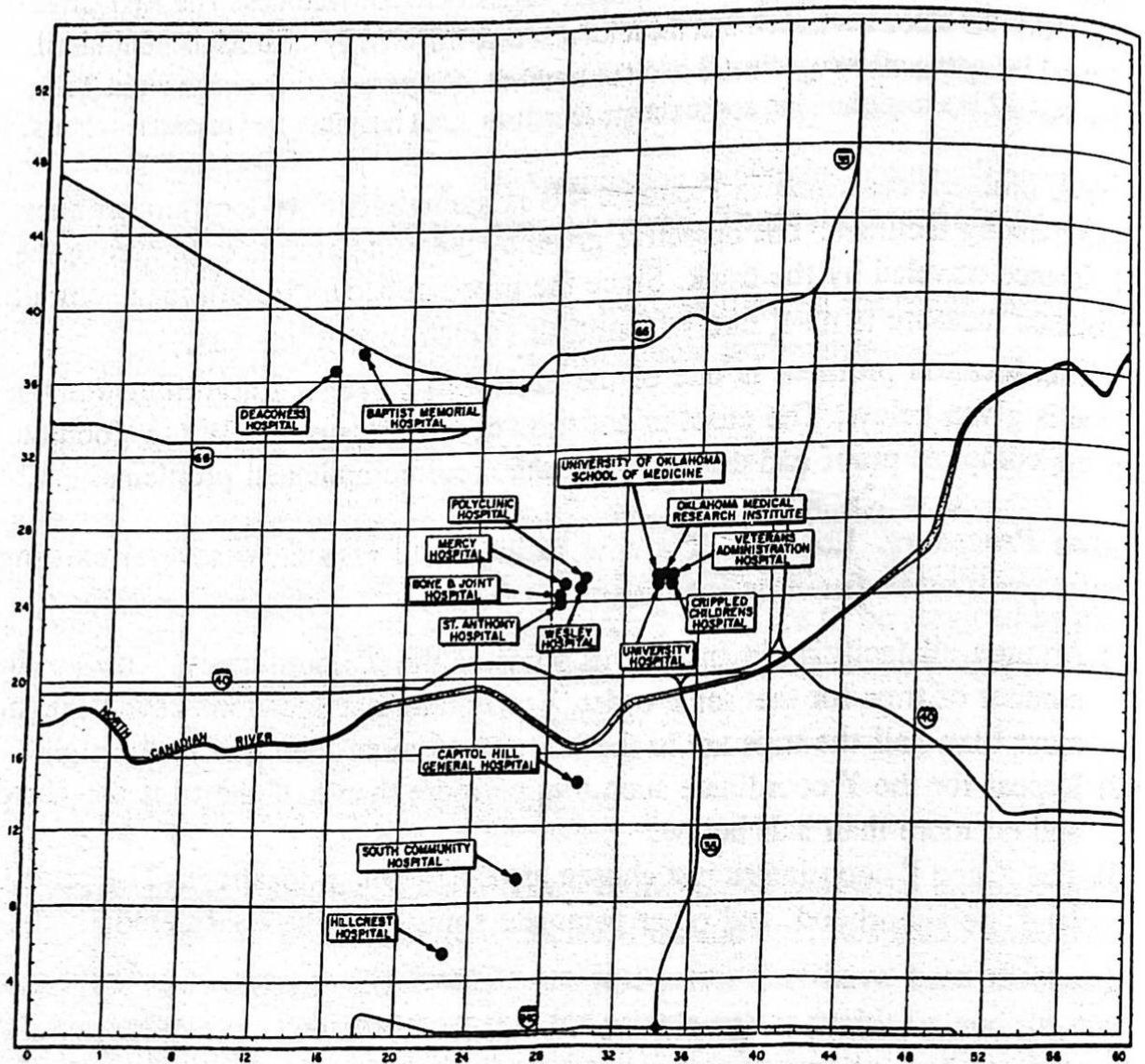


Fig. 4.12. Oklahoma city map for blood bank study.

In the example problem the calculations for the X coordinate are as follows:

	<i>X Coordinate</i>										
	<i>16.0</i>	<i>18.0</i>	<i>22.5</i>	<i>26.5</i>	<i>28.5</i>	<i>29.0</i>	<i>29.5</i>	<i>30.0</i>	<i>32.0</i>	<i>32.5</i>	<i>33.0</i>
Number of trips	5	4	10	10	4	3	6	1	4	2	1
Cumulative	5	9	19	29	33	36	42	43	47	49	50

The total number of trips is 50. The optimum x location then is that location where no more than 25 trips are to the left and no more than 25 are to the right. This occurs exactly at

$$X^* = 26.5$$

There are 19 trips to the left of 26.5 and $4 + 3 + 6 + 1 + 4 + 2 + 1 = 21$ trips to the right of 26.5.

For the Y coordinate, the same procedure is required:

	X Coordinate							
	5.5	9.5	14.0	24.0	24.5	25.0	36.5	37.5
Number of trips	10	10	5	3	6	7	5	4
Cumulative	10	20	25	28	34	41	46	50

There are 50 total trips. At

$$Y^* = 14.0$$

20 trips are to the left and exactly 25 trips are to the right of the blood bank. This implies that the optimum location is at 14, 0. At

$$Y^* = 24.0$$

however, exactly 25 trips are to the left and 22 trips are to the right of the facility. This implies that the optimum location is at 24.0.

In this example the optimal solution occurs at any point in the following range of Y values:

$$14.0 \leq Y^* \leq 24.0$$

The facility may then be located at (26.5, 14.0), (26.5, 24.0), or any point between. Of course, it is possible for the optimum location of such a facility to be a point, a line, a square, or rectangular area, depending on the details of the specific problem. In this case, the answer is a straight line since the optimal location on the X axis was exactly 26.5 but that on the Y axis could vary between 14.0 and 24.0.

As the algorithm says, the next step is to consider other factors, for example, the price of the land, the neighborhood, and cost of construction. The American Red Cross should consider all these factors before making a final decision.

4.4. Introduction to Facilities Layout

Sections 4.1 through 4.3 discussed the problem of locating facilities. Sections 4.4 through 4.7 discuss the internal layout of the facility. A layout problem is one of *locating areas*; therefore, facility location and layout are often covered in the same course and/or textbook.

The planning phase of plant layout is exceptionally important. Since an organization normally must live with the layout for a long time, any mistakes in the actual layout can be very costly. These mistakes should be made on paper long before the physical movement of equipment begins. Therefore, careful layout planning is of primary importance.

Almost always the objective in a facility layout study is to minimize total cost, but the term *total cost* is very difficult to define. Many elements comprising total cost are highly complex and obscure. For example, the following list shows just a few of these cost elements:

- (1) Construction cost.
- (2) Installation cost.
- (3) Material handling cost.
- (4) Ease of future expansion.
- (5) Production cost.
- (6) Machine downtime cost.
- (7) In-process storage cost.
- (8) Safety cost.
- (9) Ease of supervision.

Even if this list were exhaustive (which it is not), to develop a total cost function composed of these nine items would be extremely difficult.

Industrial engineers working on facility layout problems often choose as their objective the minimization of material handling cost associated with the layout. The cost of handling materials in production operations often accounts for from 30 to 95% of total production costs.

The choice of minimizing material handling cost is made for several reasons. First, material handling costs are often large and exist year after year. This is contrary to many other costs, such as construction, which occur only once or infrequently.

Second, material handling cost is easily quantifiable. Usually, the cost is proportional to distance moved, and measurement of the distances is a simple matter when given a layout of the facility.

Third, material handling cost is often that cost most affected by the layout itself. Since material handling cost is proportional to distances traveled, the layout has a pronounced effect on the amount of cost.

Because of this pronounced impact on material handling cost, it is important that the layout and material handling systems design be done simultaneously or at least with significant feedback loops. To do either without feedback to the other almost always leads to suboptimization.

Similarly, layout and resource management (especially energy and water) have a dramatic impact on each other, as will be seen in Chapter 12. A good layout will consider resource management using the principles discussed in Chapter 12.

There are two essential types of layout flow: product and process. A *product layout flow* is used when there is a large volume of one or similar products. In this the machinery or equipment is arranged so that there is a continuous flow of material in an orderly fashion throughout the plant. Paper mills, dairies, cement factories, and automotive assembly plants are good examples of product layout. The primary objective in such cases is to minimize material handling cost by properly arranging the equipment in the processing sequence. Many types of flow are possible. Figure 4.13 suggests only a few.

A *process layout flow* is needed for operations involving a small volume of many different products. This type of operation is called *job-lot manufacturing*. Since there is no mass production, there can be no smooth, continuous flow lines.

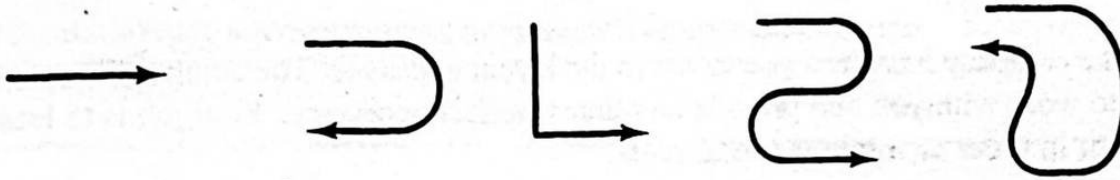


Fig. 4.13. Product flow lines.

Instead, the flow lines vary according to the particular items being produced at a particular time. In this situation it is logical to place similar machines together. For example, all lathes would be in one area, all drill presses in another, and so on.

The following example is intended to place the facility layout problem in perspective.

Example 4.4

The Whittemore Dairy of Floyd, Virginia, has decided to move its operations to a more modern plant. Remodeling and relayout of the old facility were considered but ruled out because of the age of the old building. In the new facility, four products will be produced. The four, with forecasted quantity and other general data, are shown in Table 4.4. To produce these products, the following equipment is needed:

- (1) Pasteurizer (one).
- (2) Homogenizer (one).
- (3) Pasteurization storage tank (one).
- (4) Filling machine (three) for placing products in cartons.
- (5) Cooler (one).
- (6) Laboratory equipment (one complete lab) for testing products.
- (7) By-product mixing vat and other equipment for producing cottage cheese and buttermilk.

Table 4.4 PRODUCT QUANTITY DATA FOR WHITTEMORE DAIRY

Product	Present Quantity	Growth	General
Milk	4,200 gal per day	Annual rate of 3%	Packed in $\frac{1}{2}$ -pint, 1-quart, and 5-gal boxes. Received in 4,000–6,000-gal trucks early in day—stored until used. Approximately 1 truck incoming and 20 trucks outgoing daily (all products). Normal operation is 14 hours (all products).
Juice	1,000 gal per week	Annual rate of 5% for 3 years; 3% thereafter	See <i>milk</i> for packing descriptions.
Cottage cheese	1,000 lb per week	Annual rate of 1%	Slow growth with possible phasing out. See <i>milk</i> for packing descriptions.
Buttermilk	300 gal	Annual rate of 3%	See <i>milk</i> for packing descriptions.

The company has hired you to aid in the layout endeavor. The company is completely willing to work with you and provide any data you feel necessary. Your job is to locate the equipment in order to minimize total costs.

4.5. General Considerations

Ideally, the objective in this problem is to minimize total cost, but you find that defining total cost is a very difficult task, as discussed in Section 4.4. Therefore, you assume that the objective is to minimize material handling cost by placing interacting departments as close together as possible. In an industry such as a dairy, the distances between departments may not be as important as in other industries because almost all material handling up to the packing stage is done through pipes. The longer the pipes, however, the more equipment and energy are required. Therefore, minimizing distances will minimize material handling cost.

Although not particularly true for the Whittemore Dairy, often the process used and the layout are very interdependent. That is, the choice of a particular process could dictate the total layout of the plant. In these cases, the layout is relatively simple once the process has been chosen. For the Whittemore Dairy, this is assumed not to be true. Therefore, a total layout analysis is necessary.

There are three essential types of layout: product, process, and fixed-position. *Product layout* is useful in a mass-production situation. *Process layout* is necessary in the other extreme—job-lot production. *Fixed-position layout* is useful when the product is so extremely large that it is easier to bring the workstation to the product rather than the product to the workstation. Examples of fixed-position layout include larger shipbuilding and airplane manufacturing. Here, since the ship or airplane is too large to be moved around the shop, the various stages of manufacture (particularly assembly) are performed in one place by bringing all tools to the plane or ship.

Product layout and process layout have their advantages and limitations, as demonstrated in Table 4.5. (Fixed-position layouts are seldom used, but when they are used, it is because other alternatives are infeasible.)

In the Whittemore Dairy example there are only a few products produced and almost all follow the same essential process lines (as will be seen later). For that reason, a product layout is appropriate.

Now, since the construction of the actual layout is a fairly complex problem, how do you go about doing the actual layout?

In your search for procedures you often hear systematic layout planning mentioned. Systematic layout planning is a popular approach that has proven to work well in many situations.

4.6 Systematic Layout Planning

Systematic layout planning (SLP) is an organized approach to layout planning that was developed by Richard Muther and associates. Muther's book describes the

Table 4.5 ADVANTAGES AND LIMITATIONS OF PRODUCT AND PROCESS LAYOUTS*

<i>Product Layout</i>	
Advantages	
1.	Since the layout corresponds to the sequence of operations, smooth and logical flow lines result.
2.	Since the work from one process is fed directly into the next, small in-process inventories result.
3.	Total production time per unit is short.
4.	Since the machines are located in order to minimize distances between consecutive operations, material handling is reduced.
5.	Little skill is usually required by operators at the production line; hence, training is simple, short, and inexpensive.
6.	Simple production planning and control systems are possible.
7.	Less space is occupied by work in transit and for temporary storage.
Limitations	
1.	A breakdown of one machine may lead to a complete stoppage of the line that follows a machine.
2.	Since the layout is determined by the product, a change in product design may require major alterations in the layout.
3.	The pace of production is determined by the slowest machine.
4.	Supervision is general instead of specialized.
5.	Comparatively high investment is required because identical machines (a few not fully utilized) are sometimes distributed along the line.
<i>Process Layout</i>	
Advantages	
1.	Better utilization of machines can result; consequently, fewer machines are required.
2.	A high degree of flexibility exists relative to equipment or manpower allocation for specific tasks.
3.	Comparatively low investment in machines is required.
4.	The diversity of tasks offers a more interesting and satisfying occupation for the operator.
5.	Specialized supervision is possible.
Limitations	
1.	Since longer flow lines usually result, material handling is more expensive.
2.	Production, planning, and control systems are more involved.
3.	Total production time is usually longer.
4.	Comparatively large amounts of in-process inventory result.
5.	Space and capital are tied up by work in process.
6.	Because of the diversity of the jobs in specialized departments, high grades of skill are required.

* This table is extracted with permission from Richard L. Francis and John A. White, *Facility Layout and Location: An Analytical Approach* (Englewood Cliffs, N.J.: Prentice-Hall, Inc., 1974).

procedure in detail,⁶ Francis and White discuss it in their text (footnote 1), and Figure 4.14 depicts the stages in the procedure.

As shown in Figure 4.14, all data must be gathered on current and forecasted production. In the dairy example this has already been accomplished and is presented in Table 4.4. In many studies, a list of from three to five product types will account for from 70 to 80% of the total sales volume. The remaining 20 to 30% normally

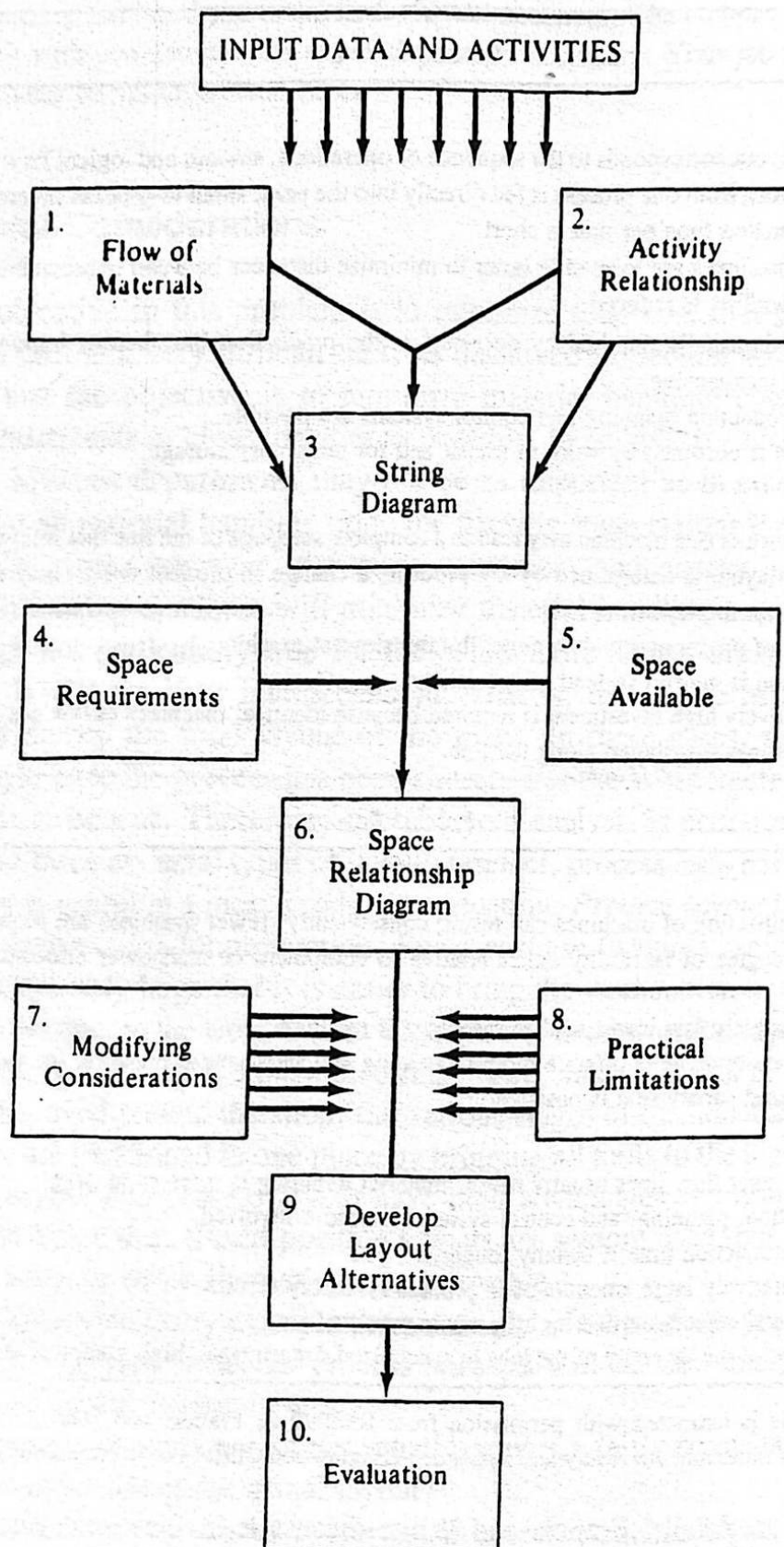


Fig. 4.14. Systematic layout planning procedure.

can be grouped so that only a few product groups need to be considered. Each product group and its respective volume for a projected horizon should be listed. The projected horizon will depend on how often the product or market changes, but usually a detailed projection for each of the next five years is adequate. Any other information that may be important to the layout should be included in the general

comments. Information such as the possible phasing out of the manufacture of cottage cheese as shown in Table 4.4 could be very critical to the long-run success of the layout.

After all product quantity data have been gathered, step 1 of the SLP procedure is the preparation of process charts graphically depicting the flow of the material through the plant. If there are just a few products, a separate operations process chart should be made for each. If there are many products, a multiproduct process chart may be used. Both are demonstrated later with the symbols shown in Figure 4.15 used to construct the charts. Actually, an operations process chart depicts only the operations and inspections. A flow process chart, which will be discussed, shows operations and inspections as well as transportations, delays, and storages.

In the operations process chart (discussed briefly in Chapter 3) a line is drawn at the upper right-hand corner of the paper reflecting the start of the largest component. Other parts of materials are fed in from the left. The sequence is from top to bottom and only operations and inspections are shown. Figure 4.16 illustrates an operations process chart for processing milk for Whittemore Dairy. A multiproduct process chart form, shown in Figure 4.17, demonstrates that the only difference is that the multiple products are listed across the top and the flow is from top to bottom only.

Sometimes additional information is needed about a particular phase of the operations. A flow process chart is used for this purpose. A flow process chart is essentially the same as an operations process chart, but a flow process chart shows more information about a specific part or item. Figure 4.18 shows a flow process chart for processing juice in the old dairy. This chart, showing storages and transportations, gives more detail than an operations process chart does. This is a highly simplified flow process chart form. Other more general forms appear in Muther's book and in almost all work simplification books.

In some situations, such as a job shop, it is difficult, if not impossible, to represent all the flow with a few charts like those discussed above. Many products

SYMBOL	MEANING
	Operation
	Transportation (Flow process chart only)
	Inspection
	Delay (Flow process chart only)
	Storage

Fig. 4.15. Process chart symbols.

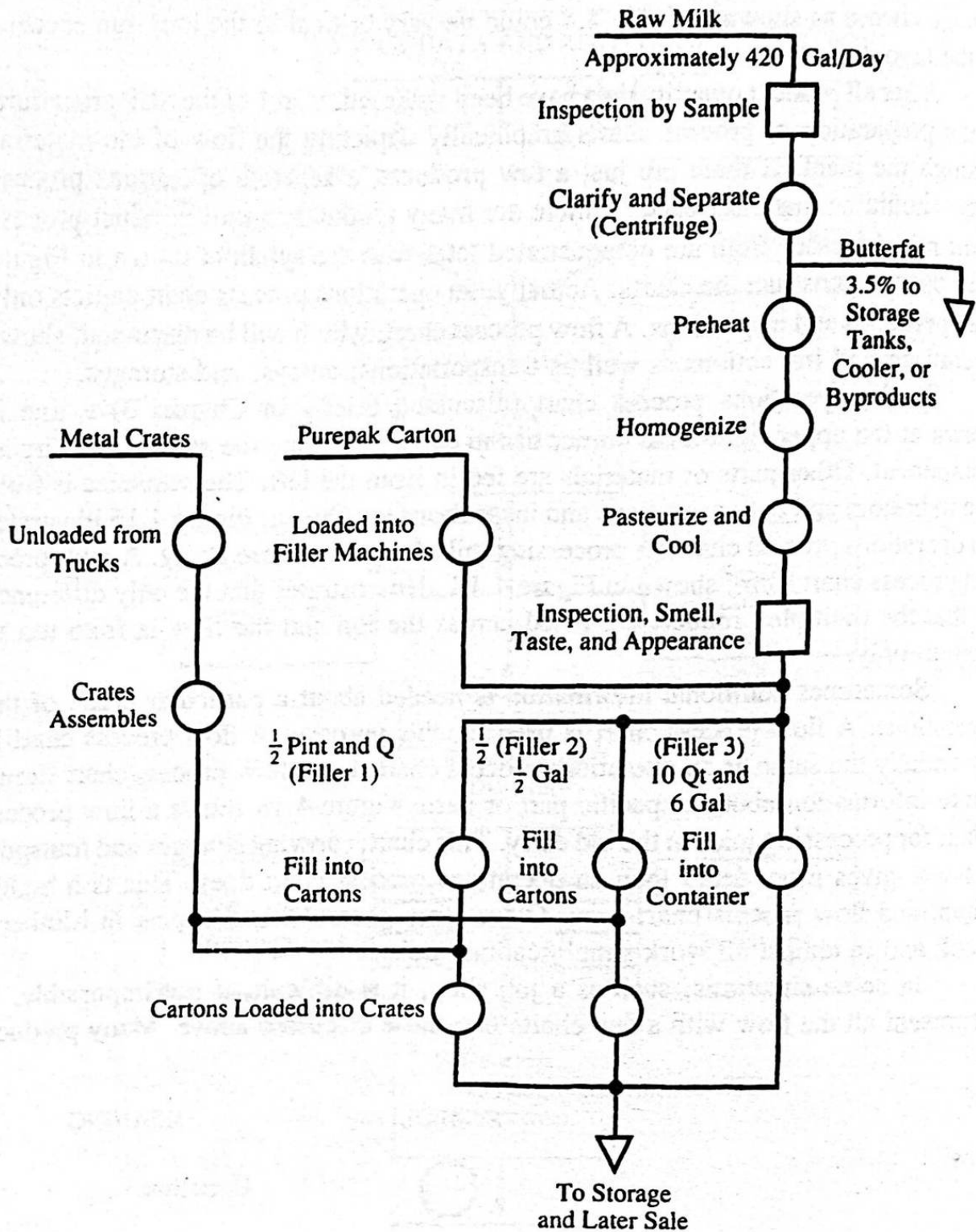


Fig. 4.16. Operations process chart depicting the processing of milk in Whittemore Dairy.

may be produced that require a large number of charts. Instead of preparing all these charts, a from-to chart may be more appropriate.

A *from-to chart* shows the number of trips from one area to another area and is based on historical data or proposed production. The trips can be weighted by product volume or some other difficulty measure if desired. Figure 4.19 shows an example of a from-to chart applied to an office situation.

Now, the flow has been depicted through one or more of the above charts;

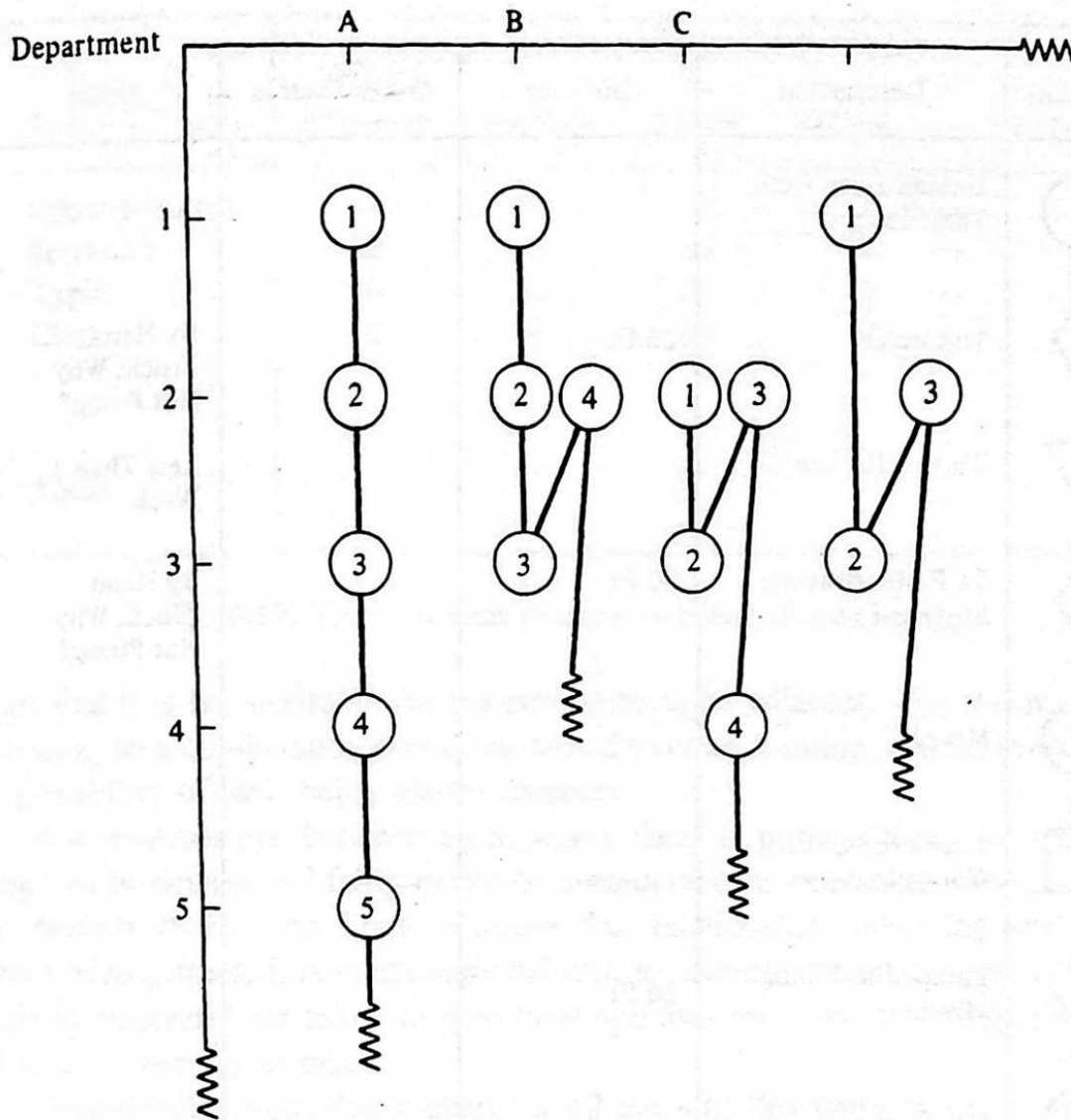


Fig. 4.17. Multiproduct process chart.

often layouts can be constructed on the basis of this flow alone. In the dairy example, almost all areas can now be laid out according to the operations process charts (Figure 4.16) or flow process charts (Figure 4.18). (Actually, several charts are necessary, one for each major product.)

In many situations, however, the flow tells only part of the story, for there are some areas that do not have any product flow and some in which the flow sequence differs for each of several products. In the dairy example the laboratory equipment, rest room, and metal crate areas do not have product flows at all. Also, the flow of cottage cheese and buttermilk is different from that of milk and juice. Therefore, you decide that more study is warranted and you find that an *activity relationship diagram* could be constructed.

Step 2 in SLP is the preparation of an *activity relationship diagram* that shows the desired closeness of departments and areas within the plant. The activity relationship diagram reflects the fact that not all important relationships can be shown by product flows. For example, it is desirable for the quality control lab in a dairy to be situated as close to the processing area as possible and for the rest room facilities to be far away from the mixing vats in the processing area.

Symbol	Description	Distance	Other Factors	Notes
	Unload Juice from Tank Trucks			
	To Cooler	25 Ft		By Hand Truck. Why Not Pump?
	Waiting for Use			Less Than 1 Week
	To Pasteurization Storage Tanks	30 Ft		By Hand Truck. Why Not Pump?
	Mixed			
	Check for Quality Control			
	Pumped to Filling Machines	20 Ft		
	Fill Cartons			
	To Cooler	20 Ft		Powered Conveyor
	Stored			Less Than 24 hr Average
	To Truck	15 Ft		Powered Conveyor
	Loaded and Shipped			

Fig. 4.18. Flow process chart for the processing of juice.

In constructing the chart all departments and areas are listed and a closeness desirability rating is given to each paired combination. Figure 4.20 is a set of ratings as proposed by Muther. Others are possible, but these seem to work very well. For any paired combination then, an A rating means that it is absolutely necessary that the two areas be located adjacent to each other. At the other extreme, an X rating

From \ To					
	Office Manager	Secretary	Typist	Clerk	... Totals
Office Manager	—	10	5	1	...
Secretary	20	—	10	10	...
Typist	10	30	—	20	...
Clerk	5	25	10	0	...
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
Totals					

Fig. 4.19. From-to chart example (number of trips per day).

means that it is not desirable for the two areas to be adjacent. The rest room and processing area combination is one that would have an X rating in order to eliminate any possibility of their being placed together.

For relationships between areas where there is product flow, the closeness rating can be determined fairly easily by considering the operations process charts, flow process charts, and from-to charts. For relationships involving one or two nonprocessing areas, it is much more difficult to determine these ratings. Often, all involved personnel are asked to give their opinions on what the ratings should be and then an average is taken.

You decide to develop a chart for all areas of this dairy and it is shown in Figure 4.21. It seems best in almost all situations to list all areas⁷ and determine the letters by considering the figures developed in the process charts and by holding consultations with the foremen, workers, and others. In this dairy example all ratings were determined either by referring to the charts whenever possible or by working with the personnel involved.

Step 3 involves using the information generated in steps 1 and 2 to prepare a string diagram showing near optimal placement of the facilities without consideration of space requirements. Figure 4.22 illustrates one possible string diagram for the dairy. The placement is done through trial and error. Normally, those areas having an A closeness are shown first and are connected with four straight lines, then E with three straight lines, and so on. When an activity has to be close to several other areas, it can be stretched out or distorted, as shown in Figure 4.22 for storage area 1. The areas may be moved around and interchanged until a final acceptable arrangement is obtained. It is helpful to visualize the straight lines as stretched rubber bands and the jagged lines as coiled springs representing varying attraction and repulsion forces. Therefore, an A rating would imply four rubber bands pulling the areas together while an I rating would imply only two rubber

⁷A typical organization can list all areas and seldom have more than 50. This seems to be an upper limit for effective representation by activity relationship diagrams.

Letter	Closeness
A	Absolutely Necessary
E	Especially Important
I	Important
O	Ordinary Closeness O.K.
U	Unimportant
X	Not Desirable

Fig. 4.20. Activity relationship diagram symbols.

Activity or Function		Activity or Function Number													
		15	14	13	12	11	10	9	8	7	6	5	4	3	2
1. Carton Storage		U	U	U	U	U	U	E	E	E	U	U	U	U	U
2. Unloading Area		U	U	U	U	U	U	U	U	U	U	U	U	I	
3. Raw Milk Storage		U	U	U	U	U	U	U	U	U	U	U	I		
4. Homogenizer		U	U	U	U	U	U	U	U	U	U	I			
5. Pasteurizer		U	U	U	U	U	U	U	U	U	I				
6. Pasteurization Storage Tanks		O	U	U	U	U	E	I	I	I					
7. Filling Machine	1	O	E	U	I	O	A	U	U						
8. Filling Machine	2	O	E	U	I	O	A	U							
9. Filling Machine	3	O	E	U	I	O	A								
10. Cooler		U	U	A	U	I									
11. Byproduct Room		O	U	U	X										
12. Restroom		U	U	O											
13. Truck Loading Area		U	U												
14. Metal Crate Loading		U													
15. Laboratory															

Fig. 4.21. Activity relationship diagram for Whittemore Dairy.

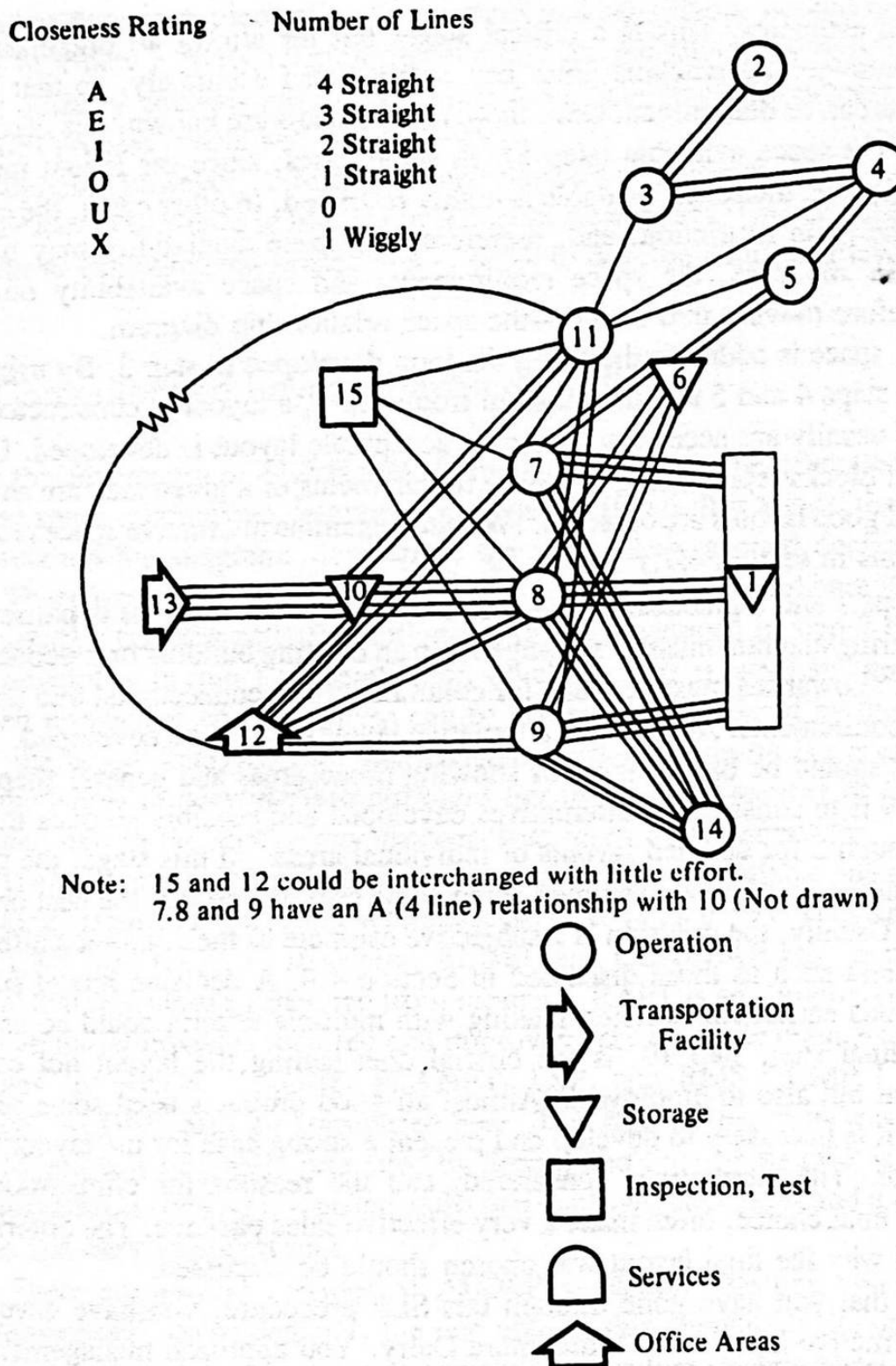


Fig. 4.22. String diagram for Whittemore Dairy.

bands. Many diagrams and arrangements will probably have to be drawn before a good layout is obtained. Normally, two or more alternatives are developed. Essentially, these alternatives each constitute a final layout. Space will have to be added and some modifications made, but the overall picture should not change much. This third step is, therefore, probably the most creative and the most important.

Step 4 may be called the *adjustment step*. Here, adjustments must be made for space needs as related to space availability; therefore, space requirements have to be determined. This can be done through calculations, adjustment of past areas,

or just good estimates. This is a critical stage, but for almost all organizations, in-process storage and machine areas can be predicted accurately, so that space requirements can be determined. Once these requirements are known, it is necessary to consider the space available (step 5). In some cases, since the layout must fit existing buildings, the space available is highly restricted. In other cases, the capital budget is the main restriction, and, therefore, the space availability may be less restricted. In any case, the space requirements and space availability must be balanced before moving into step 6—the space relationship diagram.

Here, space is added to the string diagram developed in step 3. By using the space from steps 4 and 5 and the diagram from step 3, a layout is constructed; but many trials usually are necessary before an acceptable layout is developed. Often, a number of blocks representing the space requirements of a given area are shuffled around until good layouts are obtained. (We shall examine alternative space relationship diagrams in section 4.7.)

In steps 7 and 8 practical limitations must be considered. This is particularly true if the string diagram must be massaged into an existing building or preconceived shape, or if allowances must be made for columns, utility connections, and unusual shapes or requirements. At least two alternative layouts should be developed. These alternatives should be block diagrams showing broad areas and general shapes.

Step 9 is to consider the alternatives developed and possibly go back through steps 1 through 5 for detailed layouts of individual areas. At this stage, the two or more alternatives should be examined using some criterion so that the best one can be chosen. Usually, the criterion is a subjective estimate of the combined effects of several criteria such as those discussed in Section 4.4. A decision model such as that developed earlier for decision making with multiple criteria could be useful.

The final step, step 10, is the crucial one: selling the layout not only to management but also to employees. Almost all good products need some selling. Therefore, it is necessary to develop and present a strong case for the layout to top management. The alternatives considered, and the reasons for elimination and subsequent final choice, often make a very effective sales package. The criteria and the reasons why the final layout was chosen should be discussed.

Now that you have gone through this SLP procedure, you have developed several alternative layouts for Whittemore Dairy. You approach management and make a presentation as discussed above. Management is impressed with your outstanding abilities in facility layout and accepts your final proposal.

Given that the overall layout is complete and that management agrees with the layout developed, you must now go back and lay out each department or area in detail. For example, in the Whittemore Dairy the individual pieces of test equipment must be laid out in the laboratory, various pieces of equipment must be located in the by-product processing area, and storage racks and other equipment must be located in the cooler.

To do this, you must repeat the 10-step SLP procedure for each of the departments that must be laid out in detail. Of course, some steps will be simpler than before since almost all of the data have already been gathered. For example,

in the laboratory each piece of test equipment and other items needed becomes an activity in a new activity relationship diagram for the laboratory only. Since the product quantity analysis has already been done, few new flow process charts or operations process charts will be necessary.

Suppose, however, Whittemore management tells you that they very much like your ideas but would really like to see more alternatives. One approach for you is to turn to computerized procedures in order to develop additional layouts.

4.7. Computerized Layout Planning (Optional)

One of the advantages of SLP is that alternative layouts can be generated based on different space relationship diagrams developed in step 6. It is logical that the more space relationship diagrams developed, the more likely a good decision can be made. Therefore, computerized layout packages can be a great aid here. Computerized packages take the activity relationship charts developed in step 2 or from-to charts developed in step 1 (depending on which package is used) and generate any specified number of layouts. These can be scored and compared by the computer and/or the layout analyst. The final decision must be made by the manager. *The chief advantage in computerized packages lies in the large number of alternatives generated and in the lack of any preconceived ideas by the computer.*

There are many computerized plant layout packages available and the number is growing steadily. Francis and White discuss three in their text. They are ALDEP, CORELAP, and CRAFT. Basically, ALDEP and CORELAP are "construction" types in that layouts are generated from scratch. CRAFT is an "improvement" type in that an initial layout is required and improvements are made upon it. Some other packages are RUGR, LAOPT, PLANET, LSP, and RMS COMP I. Only one package, ALDEP, will be considered here. The ALDEP program was developed by IBM and was originally presented by Seehof and Evans.⁸ ALDEP is chosen for this presentation because it was one of the first packages developed and it is relatively simple and easy to understand.

There are several versions of ALDEP available. The one you choose for Whittemore and, coincidentally, the one used in this chapter, is the one in which the first department is chosen for placement on a random selection basis. Then, the activity relationship chart (Figure 4.23, subsequently called the *Rel chart*) is scanned to find a department with a high (A or E) closeness rating with the department just placed. If one is found, that department is placed and the Rel chart is scanned for another department with a high closeness rating. If at any time, no departments can be found with a high rating, either the layout is complete or another random selection is made.

⁸J. M. Seehof and W. O. Evans, "Automated Layout Design Program," *The Journal of Industrial Engineering*, 18 (1967), 690-695.

111	455	555
111	444	555
111	444	555
112	444	555
222	333	555
222	333	555
222	333	555

Layout

	12345
1	S
2	AS
3	IES
4	UOUS
5	IUUES

Rel Chart

S Implies "Same"

Fig. 4.23. Sample rel chart and layout for ALDEP program.

An important part of the ALDEP program is the scoring of alternative layouts. For this scoring, the program considers immediate adjacencies only and assigns the following values to those adjacencies:

$$A = 64 \quad O = 1$$

$$E = 16 \quad U = 0$$

$$I = 4 \quad X = -1,024$$

For example, the simple layout and Rel chart shown in Figure 4.23 has the following immediate adjacencies and scores:

Departments	Score	Departments	Score
1-4	0	3-5	0
1-2	64	4-5	16
2-3	16	3-4	0
2-4	1		

The total points are

$$(0 + 64 + 16 + 1 + 0 + 16 + 0)2 = 194$$

The number of points must be multiplied by 2 since the program considers two-way adjacency (i.e., a 1-4 relationship is also a 4-1 relationship).

ALDEP uses a vertical scanning method of locating departments by laying a vertical strip of a width specified by the user across the depth of the layout. Then the strip is turned around and the process is repeated. As the strip is being laid, the departments are being placed. In the example shown in Figure 4.23 the scanning width is 3 and the first department located is department 1. The others are located in the order 2, 3, 4, 5.

As an example of the use of ALDEP, the Rel chart in Figure 4.21 is to be used as input to the program to lay out the Whittemore Dairy.

The basic inputs to the ALDEP program are the following:

- (1) The vertical scanning width. Here a width of 3 is used, but by using other values, different layouts can be obtained.

- (2) The Rel chart. The chart shown in Figure 4.21 is used.
- (3) The departmental area requirements. These values are shown in Figure 4.24. You developed these by working with Whittemore management and other dairies.

It is also necessary to tell the program what minimum score is acceptable. In this case, a minimum score of 0 is used for the first run in which 20 layouts are done. For the second run, the best score of the first 20 is used as minimum and 20 more are run. The normal procedure is to continue this until either no additional layouts are developed or else a decision is made.

You feed all this into the program and develop the first set of 20 runs. The first layout developed (which is probably not the best) is reproduced in Figure 4.25. The score for this layout is 188, developed by looking at the following relationships:

<i>Relationships</i>	<i>Points</i>	<i>Relationships</i>	<i>Points</i>	
5-10	0	4-9	0	
10-7	64	9-8	0	
7-1	16	8-6	4	
1-11	0	8-15	1	
1-14	0	8-12	4	
1-3	0	6-2	0	
14-11	0	6-13	0	
14-3	0	6-12	0	
3-11	1	12-13	1	$94 \times 2 = 188$
11-4	1	2-13	0	
11-9	1	6-15	1	
4-6	0	15-12	0	
4-8	0	Total	94	

<i>Department</i>	<i>Area (Ft²)</i>	<i>Department</i>	<i>Area (Ft²)</i>
1 Carton Storage	100	9 Filling Machine 3	50
2 Unloading Area	50	10 Cooler	300
3 Raw Milk Storage	100	11 Byproduct Room	300
4 Homogenizer	50	12 Restroom	30
5 Pasteurizer	50	13 Truck Loading Area	40
6 Pasteurization Storage Tank	90	14 Metal Crate Loading	40
7 Filling Machine 1	100	15 Lab	50
8 Filling Machine 2	100		

Fig. 4.24. Departmental area requirements for Whittemore Dairy.

TRIAL LAYOUT 18 SCORE = 100

[illegible]

UNDER THE RULES FOR EVALUATION, THIS LAYOUT DOES NOT SATISFY THE FOLLOWING NECESSARY CLOSE RELATIONSHIPS

108	110
109	110
110	108
110	109
110	113
113	110

Fig. 4.25. First ALDEP layout of Whittemore Dairy.

The seventeenth layout (reproduced in Figure 4.26) produces the maximum score, which is 440. This layout has some interesting properties. First, at the extreme left end of the building is the cooler (number 10) with a truck loading area (number 13) for moving material from the cooler to the trucks. This looks good and would work very well for Whittemore Dairy. Also, the product *essentially* starts from the right and flows to the left, which is good.

Finally, the layout brings some interesting points to light. The management always thought that the filling machines should be located together. Yet, neither the layout nor the Rel chart showed this; consequently, management agreed that this was not necessary (the computer is not constrained by any mental blocks or narrow points of view). The laboratory is located in the middle of the building, which is good, but so is the metal crate unloading area, which is bad. You and Whittemore management decide to generate some more layouts utilizing the best parts of this one.

One of the advantages of ALDEP is that areas can be preassigned or fixed.

[illegible]

108	110
110	108

Therefore, you decide to preassign (number 13) exactly as it is and to preassign the unloading area (number 2) to the upper right-hand corner of the building. Since both would be at the top side of the building, road construction would be facilitated.

You generate 20 more runs. One of those runs scores 512 points and seems to offer potential. You subsequently modify it (to round off corners, and so on) and present it to Whittemore management. The semifinished product is shown in Figure 4.27.

Management likes several things particularly about the layout represented in Figure 4.27. The flow moves from right to left without much back-tracking. The unloading, metal crate, truck loading, and carton storage areas are all located on one side of the building, which facilitates road construction. The by-product room is located directly in the middle of the plant, which facilitates whatever manual material handling there is. Management accepts your proposal, rewards you with a fat bonus, and asks you to do the detailed layout.

The decision on the overall layout has been made. You must now go back and repeat the SLP procedure for each of the departments and lay them out in detail as discussed in Section 4.6.

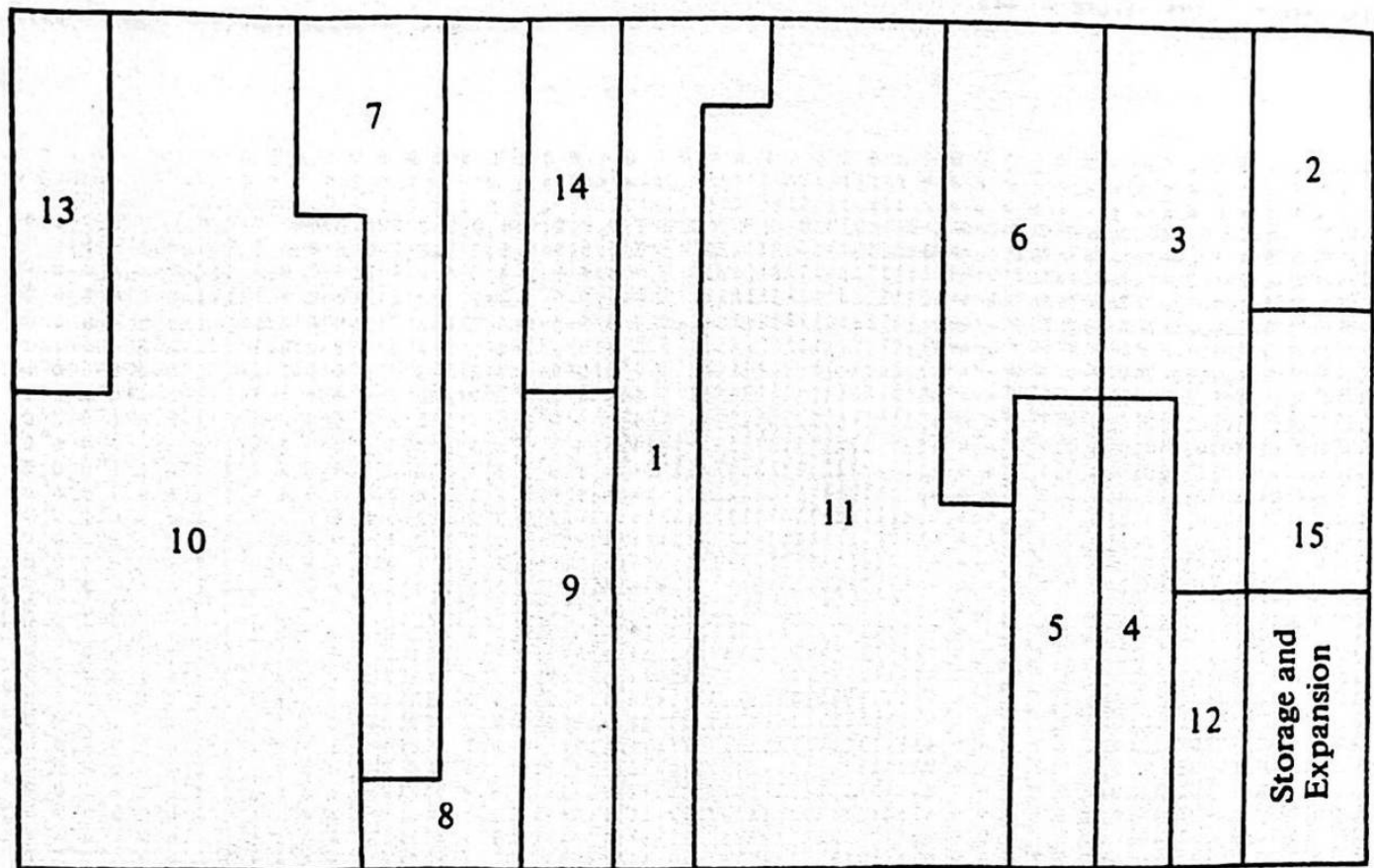


Fig. 4.27. Final Whittemore Dairy layout.

4.8. Impact of Computers

Computers have been used in various areas of location-layout studies with mixed successes. Undeniably, the computer has been immensely useful in storing and manipulating large amounts of data as might be required in location decisions and in manipulating large mathematical models.

Also, computer graphics can be a big aid in location and layout studies. Alternative locations or layouts can be generated and displayed on cathode-ray tubes (CRTs) or drawn in hard copy. Some graphics routines allow manual intervention with a light pen or mouse and rescoring. This can be very helpful and it can mean the difference between a worthless computer exercise and a realistically feasible location or layout. Recently, computer-aided design (CAD) packages have made design of alternatives so easy and the product so professional looking that layout planners often "sketch" many alternatives and seek to improve them. Thus the scope of computerized layout planning is changing.

Finally, research is under way to impart more realism and architectural considerations into layout computer programs with some interesting successes. Future computer utilization in this area seems to be toward interactive routines capable of recognizing realistic opportunities.